BUILDING PLANNING AND DRAWING

(23A01304)

B.Tech. Civil Engineering - II Year I Semester - JNTUA R23

L-T-P-C : 0-1-2-2

Course Objectives

- Initiating the student to different building bye-laws and regulations.
- Imparting the planning aspects of residential buildings and public buildings.
- Giving training exercises on various signs and bonds.
- Giving training exercises on different building units.
- Imparting the skills and methods of planning of various buildings.

Course Outcomes

- Plan various buildings as per the building by-laws.
- Distinguish the relation between the plan, elevation and cross section and identify the form and functions among the buildings.
- Draw signs and bonds.
- Draw different building units.
- Learn the skills of drawing building elements and plan the buildings as per requirements.

Syllabus (Drawing Exercises)

1. Detailing & Drawing of Sign Conventions
2. Detailing & Drawing of English Bond
3. Detailing & Drawing of Flemish Bond
4. Detailing & Drawing of Doors
5. Detailing & Drawing of Windows
6. Detailing & Drawing of Ventilators & Roofs
7. Drawing of Line Diagram of Residential Buildings by using Building Bye-Laws
8. Drawing of Plan, Elevation & Section from line diagram for a single Storey Building
9. Drawing of Plan, Elevation & Section for Hospital Building
10. Drawing of Plan, Elevation & Section for Industrial Building

Text Books

1. Planning, designing and Scheduling, Gurcharan Singh and Jagdish Singh
2. Building planning and drawing by M. Chakraborti.
3. Building drawing, M G Shah, C M Kale and S Y Patki, Tata McGraw Hill, New Delhi.

Reference Books

1. National Building Code 2016 (Volume I & II).
2. Principles of Building Drawing, M G Shah and C M Kale, Trinity Publications, New Delhi.
3. Civil Engineering drawing and House planning, B. P. Verma, Khanna publishers, New Delhi.

Exercise 1: Sign Conventions

Aim:

To draw the conventional signs and symbols used in building drawing for different materials.

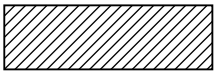
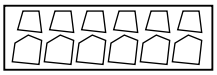
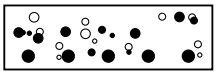
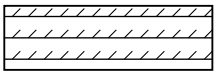
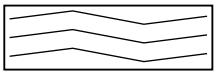
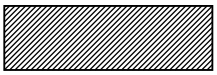
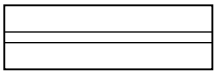
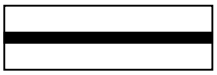
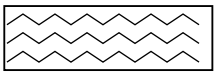
Materials / Instruments:

Drawing sheet (A2), drawing board, T-square, set squares (45 and 30-60 degree), compass, divider, scale (1:20 / 1:50 / 1:100), pencils (H, HB, 2H), eraser, drawing clips.

Procedure:

1. Fix the drawing sheet on the board and draw the border and title block.
2. Divide the sheet into equal rectangles of 40 mm x 15 mm.
3. Draw the hatching / symbol for each material inside its rectangle as per IS 962.
4. Write the name of the material below each symbol.
5. Draw the thick outlines and complete the sheet.

Observations (Conventions drawn):

S.No.	Material	Conventional sign	Remark
1	Brick masonry		Section hatched at 45 degrees
2	Stone masonry		Irregular stones with joints
3	Plain cement concrete (PCC)		Dots and small circles
4	Reinforced cement concrete (RCC)		Concrete with bars shown as dots
5	Earth / earth filling		Horizontal lines with short slants
6	Sand		Fine dots
7	Timber (wood)		Wavy grain lines
8	Steel		Close 45 degree lines
9	Glass		Two thin parallel lines
10	Damp proof course (DPC)		Thick black line
11	Water		Wavy lines

Result:

The conventional signs for 11 building materials were drawn as per IS 962.

Exercise 2: English Bond

Aim:

To draw the elevation and plan of alternate courses of a one-brick thick wall in English bond.

Materials / Instruments:

Drawing sheet (A2), drawing board, T-square, set squares (45 and 30-60 degree), compass, divider, scale (1:20 / 1:50 / 1:100), pencils (H, HB, 2H), eraser, drawing clips.

Procedure:

1. Take the size of the brick as 190 x 90 x 90 mm (nominal size with mortar 200 x 100 x 100 mm).
2. Draw a wall length of 1400 mm and 8 courses (each 100 mm high) in elevation at 1:5 scale.
3. In the first (stretcher) course draw all bricks as stretchers (length 200 mm).
4. In the second (header) course draw a queen closer (50 mm) after the end header, then all headers (100 mm).
5. Repeat the two courses alternately so that vertical joints do not come over each other (lap = 1/4 brick).
6. Draw the plan of the first two courses below the elevation and shade the closers.

Observations:

Item	Value
Brick size (actual / nominal)	190 x 90 x 90 mm / 200 x 100 x 100 mm
Wall thickness	1 brick = 200 mm (nominal)
Wall length shown	1400 mm
Courses (height)	8 (4 stretcher + 4 header), 100 mm each
Lap	1/4 brick (50 mm) using queen closer
Bricks per m^3	500 (approx.)

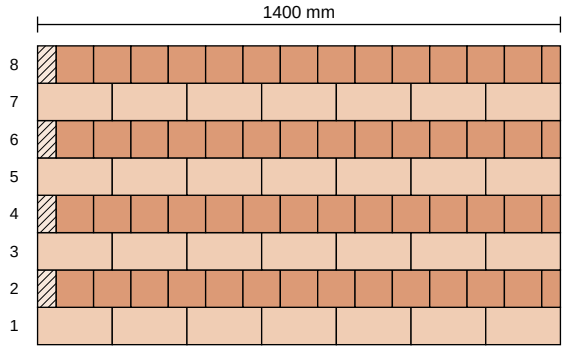


Fig: English bond - elevation (courses 1 to 8, bottom to top)

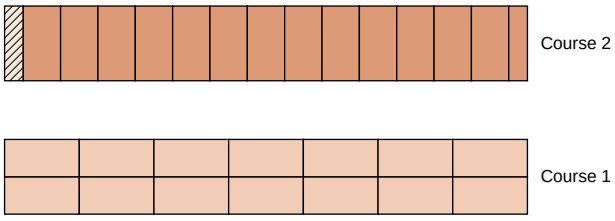


Fig: English bond - plan of alternate courses (hatched = queen closer)

Result:

English bond was drawn. Stretcher and header courses alternate; the queen closer next to the quoin header gives the necessary lap.

Exercise 3: Flemish Bond

Aim:

To draw the elevation and plan of alternate courses of a one-brick thick wall in Flemish bond.

Materials / Instruments:

Drawing sheet (A2), drawing board, T-square, set squares (45 and 30-60 degree), compass, divider, scale (1:20 / 1:50 / 1:100), pencils (H, HB, 2H), eraser, drawing clips.

Procedure:

1. Take the brick size as 190 x 90 x 90 mm (nominal 200 x 100 x 100 mm).
2. Draw 8 courses of 100 mm each over a wall length of 1400 mm at 1:5 scale.
3. In every course draw a header (100 mm) and a stretcher (200 mm) alternately.
4. In alternate courses start with a 3/4 bat (150 mm) so that each header lies in the centre of the stretcher above or below it.
5. Draw the plan of two successive courses and hatch the bats.

Observations:

Item	Value
Brick size (actual / nominal)	190 x 90 x 90 mm / 200 x 100 x 100 mm
Wall thickness	1 brick = 200 mm (nominal)
Pattern in each course	Header (100) + Stretcher (200), repeated every 300 mm
Bat at start of alternate course	3/4 bat = 150 mm, course height 100 mm
Appearance	Better than English bond, slightly weaker

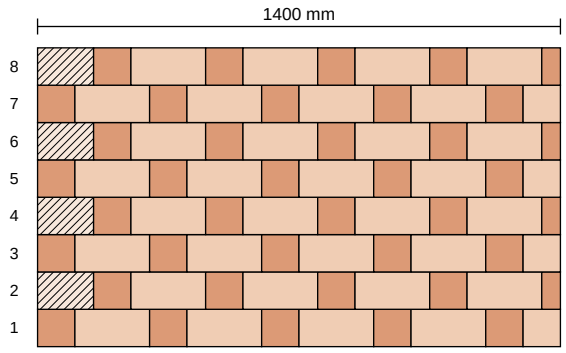


Fig: Flemish bond - elevation (courses 1 to 8, bottom to top)

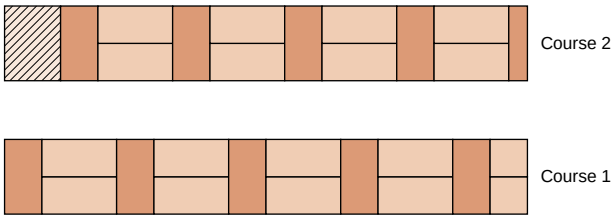


Fig: Flemish bond - plan of alternate courses (hatched = 3/4 bat)

Result:

Flemish bond was drawn with alternate headers and stretchers in every course, headers centred on stretchers above and below.

Exercise 4: Doors

Aim:

To draw the elevation of a four-panelled single-shutter wooden door with its frame.

Materials / Instruments:

Drawing sheet (A2), drawing board, T-square, set squares (45 and 30-60 degree), compass, divider, scale (1:20 / 1:50 / 1:100), pencils (H, HB, 2H), eraser, drawing clips.

Procedure:

1. Draw the outer frame 1150 mm x 2100 mm at 1:10 scale.
2. Draw the shutter 1000 mm x 2025 mm inside the frame.
3. Mark the top rail (100 mm), lock rail (150 mm at 1000 mm height), bottom rail (200 mm) and the stiles (100 mm).
4. Draw the central muntin (100 mm) and the four panels with 16 mm chamfer lines.
5. Give the dimensions and write the names of the parts.

Observations:

Part	Size (mm)	Material
Door frame (outer size)	1150 x 2100	Teak / sal wood, 100 x 75 sections
Shutter	1000 x 2025 x 35	Seasoned wood
Stiles	100 x 35	Wood
Top rail	100 x 35	Wood
Lock rail	150 x 35	Wood
Bottom rail	200 x 35	Wood
Muntin	100 x 35	Wood
Panels (4 nos.)	approx. 350 x 750 (upper), 350 x 650 (lower), 20 thick	Plywood / wood

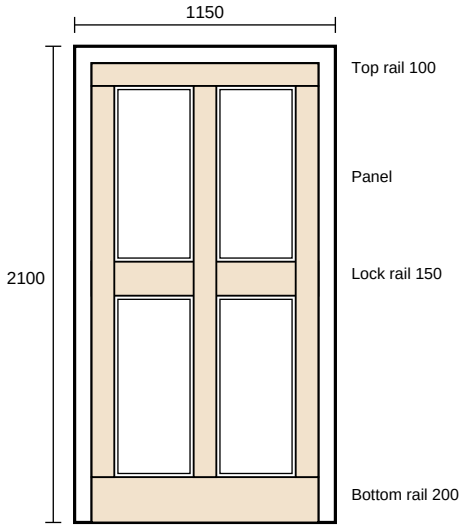


Fig: Four-panelled door - front elevation (scale 1:10)

Result:

Elevation of a four-panelled door of clear opening 1000 mm x 2025 mm was drawn with all component sizes.

Exercise 5: Windows

Aim:

To draw the elevation of a two-shutter glazed wooden casement window.

Materials / Instruments:

Drawing sheet (A2), drawing board, T-square, set squares (45 and 30-60 degree), compass, divider, scale (1:20 / 1:50 / 1:100), pencils (H, HB, 2H), eraser, drawing clips.

Procedure:

1. Draw the outer frame 1300 mm x 1300 mm at 1:10 scale with the sill below.
2. Mark the clear opening 1200 mm x 1200 mm.
3. Divide the width into two equal shutters of 600 mm each.
4. Provide top, middle and bottom rails and stiles of 50 mm width and draw two glass panes in each shutter.
5. Add the dimensions and the names of the parts.

Observations:

Part	Size (mm)	Material
Window frame	1300 x 1300 (outer)	Wood 100 x 75
Clear opening	1200 x 1200	-
Shutters (2 nos.)	600 x 1200 x 35	Wood
Rails and stiles	50 x 35	Wood
Glass panes (4 nos.)	approx. 480 x 520 x 3	Clear glass
Sill	1300 x 150	Stone / precast concrete
Lintel	1500 x 230 x 150	RCC

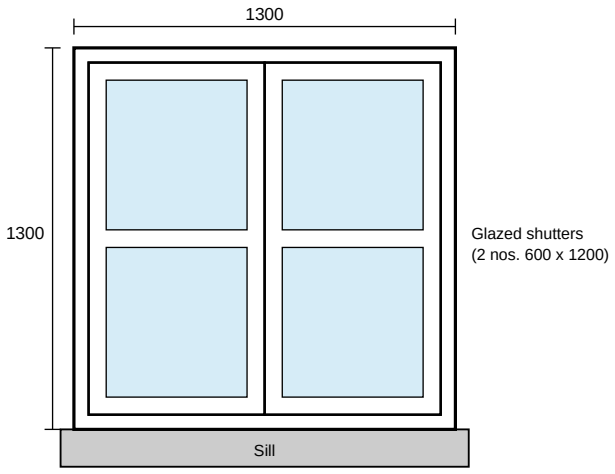


Fig: Casement window - front elevation (scale 1:10)

Result:

Elevation of a two-shutter glazed window of 1200 mm x 1200 mm clear opening was drawn.

Exercise 6: Ventilators and Roofs

Aim:

To draw a fixed-louver glass ventilator and a king post roof truss of 6 m span.

Materials / Instruments:

Drawing sheet (A2), drawing board, T-square, set squares (45 and 30-60 degree), compass, divider, scale (1:20 / 1:50 / 1:100), pencils (H, HB, 2H), eraser, drawing clips.

Procedure:

1. Draw the ventilator frame 600 mm x 450 mm and divide the height into four equal louvers.
2. Draw glass louvers inclined at 45 degrees and show the dimensions.
3. Take the truss span as 6000 mm and rise as 1500 mm (pitch 1 in 4).
4. Draw the tie beam, two principal rafters and the king post.
5. Draw the struts from the king post foot to the rafters and place purlins at the strut ends.
6. Mark the wall plates and the ridge and write the names of the members.

Observations:

Item	Size / value
Ventilator frame	600 x 450 mm (wood 75 x 50)
Louvers	4 nos., glass 600 x 100 x 5 mm
Truss span	6000 mm
Rise / pitch	1500 mm / 1 in 4 (26.6 degrees)
Tie beam	6000 x 150 x 75 mm
Principal rafters (2 nos.)	3354 x 100 x 75 mm
King post	1500 x 100 x 75 mm
Struts (2 nos.)	approx. 1530 x 75 x 50 mm
Purlins	75 x 100 mm

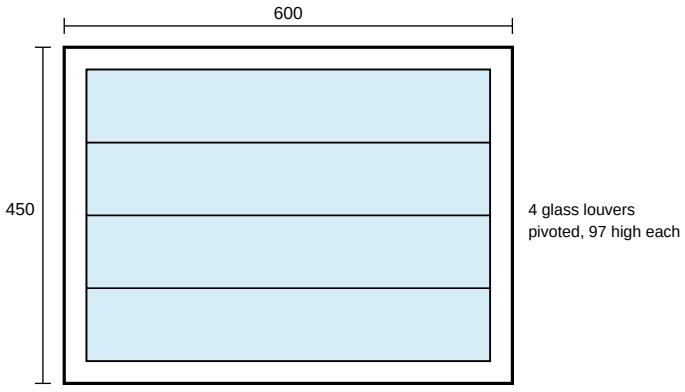


Fig: Louvered ventilator - front elevation

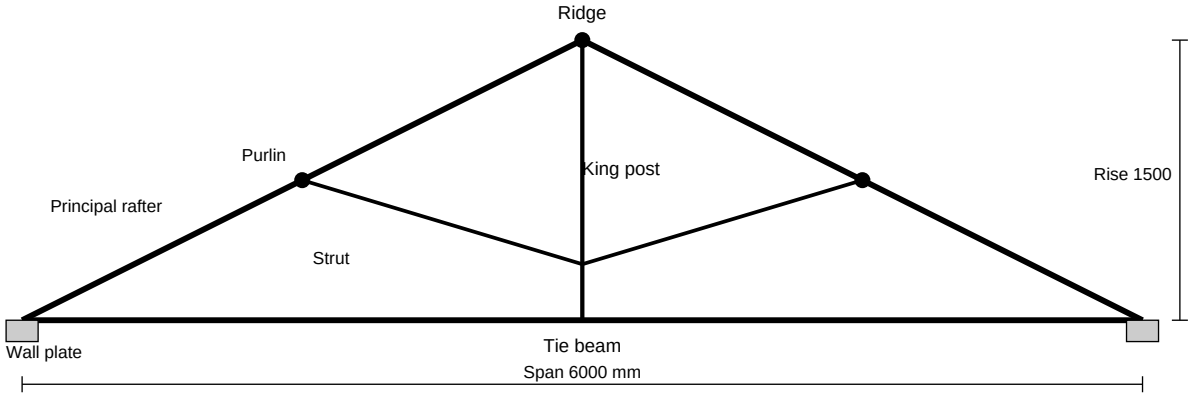


Fig: King post roof truss (span 6 m)

Result:

A 600 x 450 mm glass-louvered ventilator and a 6 m span king post roof truss with 1500 mm rise were drawn.

Exercise 7: Line Diagram of a Residential Building

Aim:

To prepare the line diagram of a single-storey residential building (2 bed room) as per building bye-laws.

Materials / Instruments:

Drawing sheet, pencils, scale 1:100, set squares, eraser. Reference: NBC 2016 and local municipal bye-laws.

Procedure:

1. Take plot size 15.0 m x 13.0 m and leave the required set-backs (front 3.0 m, rear 2.0 m, sides 3.0 m).
2. Take the built-up size as 9.0 m x 8.0 m on centre lines.
3. Arrange the rooms: verandah in front, living room and bed room 2 behind it, then bed room 1, kitchen, store and bath + WC at the rear.
4. Check each room against the minimum area and width required by the bye-laws.
5. Draw the line diagram at 1:100 scale and write the room names and sizes.

Observations:

Site data: Plot = 15.0 x 13.0 m = 195 m^2; built-up area = 9.0 x 8.0 = 72 m^2; plot coverage = 36.9 % (permissible about 60 %); set-backs: front 3.0 m, rear 2.0 m, sides 3.0 m each; height of ceiling = 3.0 m.

Room	Size provided (m)	Area provided (m^2)	Min. area required (m^2)	Min. width required (m)	Remark
Verandah	9.0 x 1.5	13.5	-	1.5	Satisfied
Living room	5.4 x 3.0	16.2	9.5	2.4	Satisfied
Bed room 2	3.6 x 3.0	10.8	9.5	2.4	Satisfied
Bed room 1	3.6 x 3.5	12.6	9.5	2.4	Satisfied
Kitchen	3.0 x 3.5	10.5	5.0	1.8	Satisfied
Store	2.4 x 1.7	4.1	-	-	Satisfied
Bath + WC	2.4 x 1.8	4.3	2.8	1.2	Satisfied

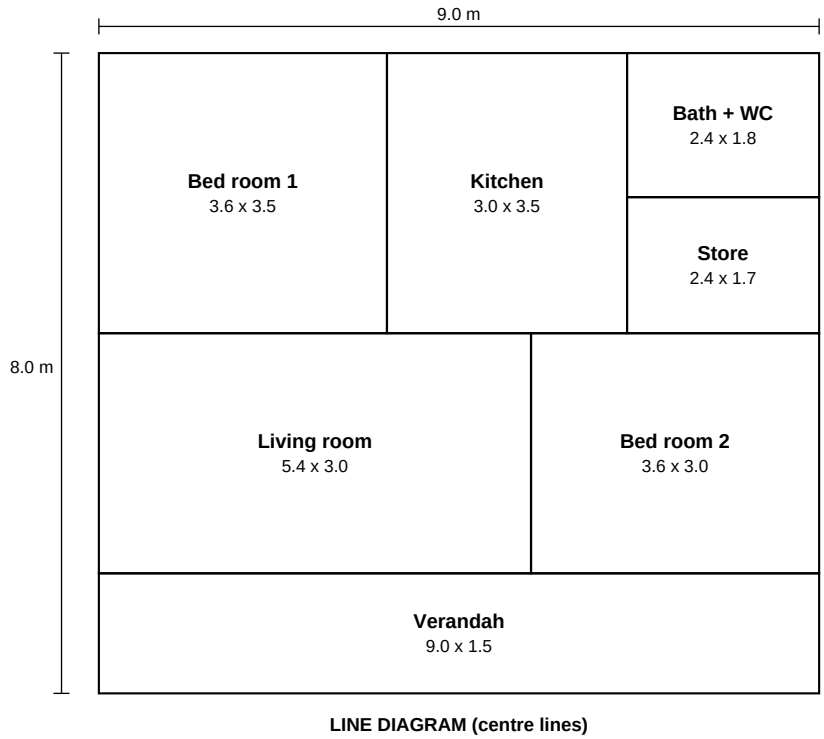


Fig: Line diagram of 2-bed room residential building (scale 1:100)

Result:

The line diagram of the residential building was prepared and all rooms satisfy the minimum size requirements of the bye-laws.

Exercise 8: Plan, Elevation and Section of a Single Storey Building

Aim:

To draw the plan, front elevation and a section of the single-storey building from the line diagram of Exercise 7.

Materials / Instruments:

Drawing sheet, pencils, scale 1:100, set squares, compass, eraser.

Procedure:

1. Convert the line diagram into a plan by giving 230 mm thick walls on the centre lines.
2. Mark the doors, windows and ventilators as per the schedule of openings.
3. Draw the front elevation by projecting up from the plan: plinth 0.45 m, wall height 3.0 m, slab 150 mm and parapet 600 mm.
4. Draw the section A-A through the verandah, living room and bed room 1 showing foundation, plinth, floor, lintels and roof slab.
5. Mark the dimensions, room names and the title block.

Observations (Schedule of openings):

Mark	Description	Size (m)	Nos.
D1	Main door (wood, panelled)	1.00 x 2.10	1
D2	Internal door	0.90 x 2.10	4
D3	Bath / WC door	0.75 x 2.10	1
W1	Window (sill 0.9 m)	1.20 x 1.20	5
W2	Window (sill 0.9 m)	1.00 x 1.20	1
V	Ventilator	0.60 x 0.45	1

Other data: foundation 0.9 m wide, 1.0 m deep with PCC 1:5:10; superstructure in brick masonry 230 mm thick; RCC slab 150 mm thick; floor level 0.45 m above GL.

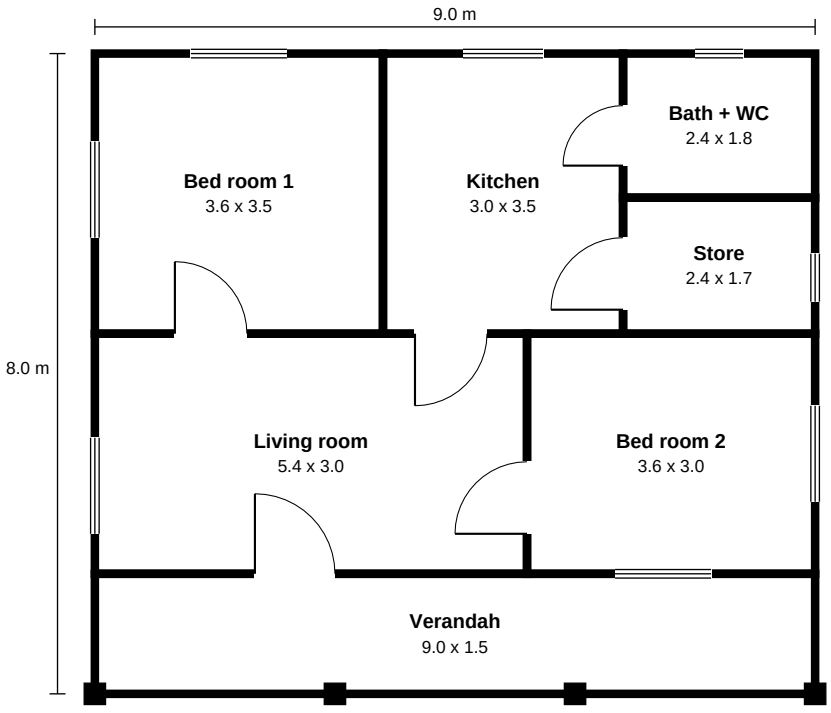


Fig: Plan of the single-storey building (scale 1:100)

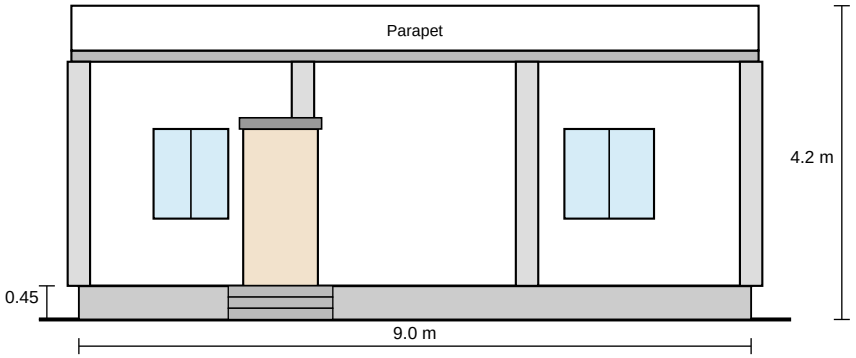


Fig: Front elevation

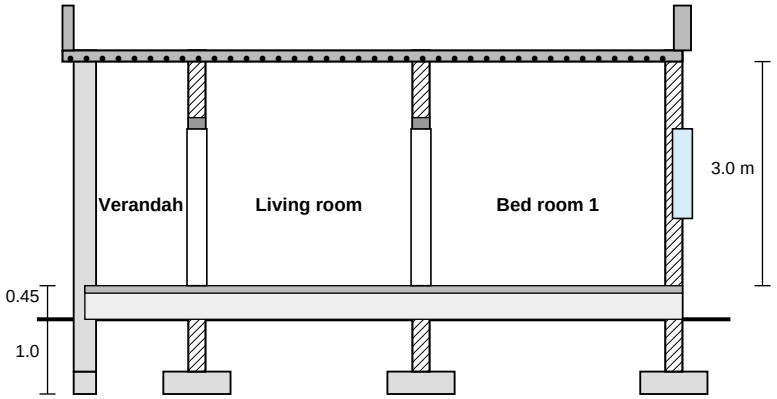


Fig: Section A-A

Result:

Plan, elevation and section of the single-storey building (plinth area 72 m^2) were drawn from the line diagram.

Exercise 9: Plan, Elevation and Section of a Hospital Building

Aim:

To draw the plan, elevation and section of a 24 m x 14 m two-storey hospital block (ground floor plan).

Materials / Instruments:

Drawing sheet, pencils, scale 1:100, set squares, eraser.

Procedure:

1. Provide a central corridor 2.4 m wide along the length of the block.
2. Arrange reception, OPD rooms, pharmacy, laboratory and X-ray room on one side of the corridor on the ground floor.
3. Arrange general wards, operation theatre and nurses station on the other side.
4. Provide the main entrance at the centre and draw the elevation with windows at each floor.
5. Draw the section A-A across the corridor showing floors, slabs and windows.

Observations:

Room / Unit	Size (m)	Area (m²)
Reception / Waiting	6.0 x 5.8	34.8
OPD 1	4.0 x 5.8	23.2
OPD 2	4.0 x 5.8	23.2
Pharmacy	4.0 x 5.8	23.2
Lab	3.0 x 5.8	17.4
X-ray	3.0 x 5.8	17.4
Corridor (2.4 m)	24.0 x 2.4	57.6
General Ward A (8 beds)	8.0 x 5.8	46.4
General Ward B (8 beds)	8.0 x 5.8	46.4
Operation Theatre	4.5 x 5.8	26.1
Nurses station	3.5 x 5.8	20.3

Total built-up area on ground floor = 24.0 x 14.0 = 336 m^2; storey height = 3.6 m; plinth = 0.6 m; wall thickness = 230 mm; each ward provides about 5.8 m^2 per bed.

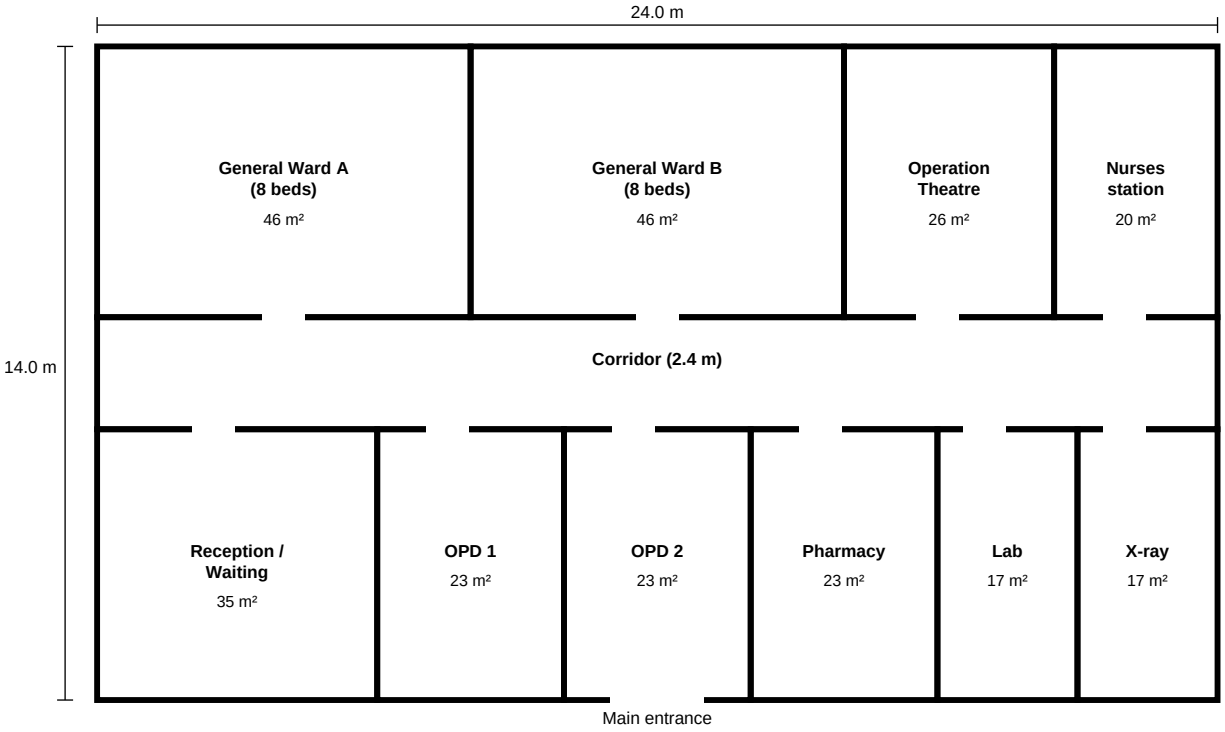


Fig: Ground floor plan of the hospital (scale 1:100)

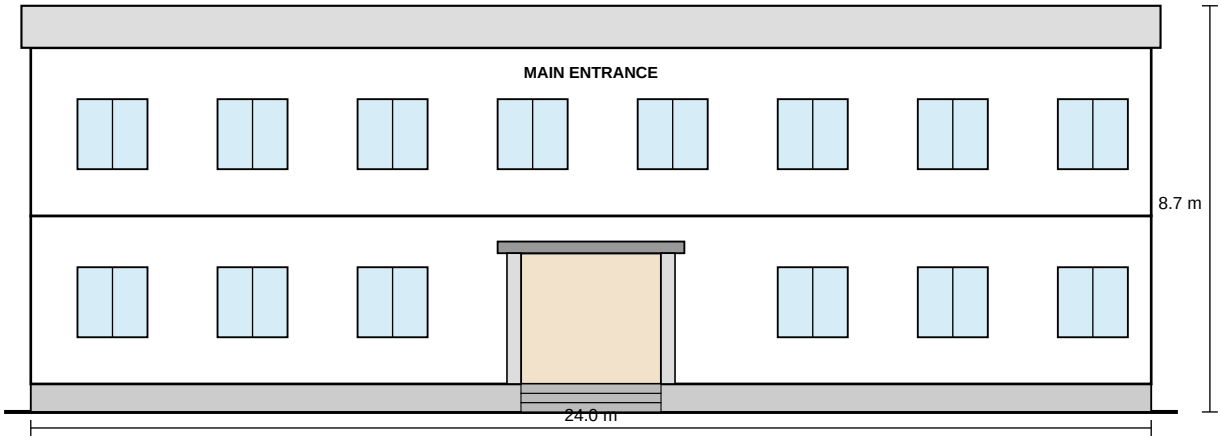


Fig: Front elevation (G + 1)

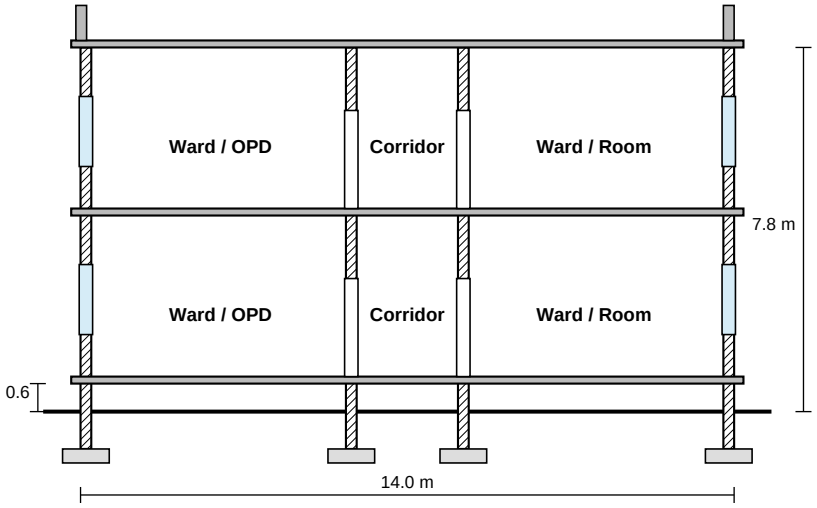


Fig: Section A-A across the corridor

Result:

Plan, elevation and section of the hospital block were drawn with the required functional arrangement of rooms.

Exercise 10: Plan, Elevation and Section of an Industrial Building

Aim:

To draw the plan, elevation and section of an industrial building (steel truss shed) of size 24 m x 12 m.

Materials / Instruments:

Drawing sheet, pencils, scale 1:100, set squares, eraser.

Procedure:

1. Take 6 bays of 4.0 m each (total length 24.0 m) and a span of 12.0 m.
2. Provide columns at every bay on both sides and an office and store block of 4.0 m x 6.0 m each at one end.
3. Provide a rolling shutter of 4.0 m x 4.5 m at the other end and a side door.
4. Draw the end elevation with brick wall up to 1.2 m and sheeting above; eaves height 6.0 m and ridge height 8.0 m.
5. Draw section A-A showing the steel truss, columns and foundations.

Observations:

Item	Size / value
Overall size	24.0 m x 12.0 m = 288 m^2
Production hall	20.0 m x 12.0 m = 240 m^2
Office	4.0 m x 6.0 m = 24 m^2
Store	4.0 m x 6.0 m = 24 m^2
Bay spacing	4.0 m (6 bays)
Eaves height / ridge height	6.0 m / 8.0 m
Roof	Steel truss, rise 2.0 m (1 in 6), GI sheets
Columns	ISMB 300 at 4.0 m c/c, isolated footings 1.5 x 1.5 m
Rolling shutter	4.0 x 4.5 m
Windows	2.0 x 1.5 m in the end wall

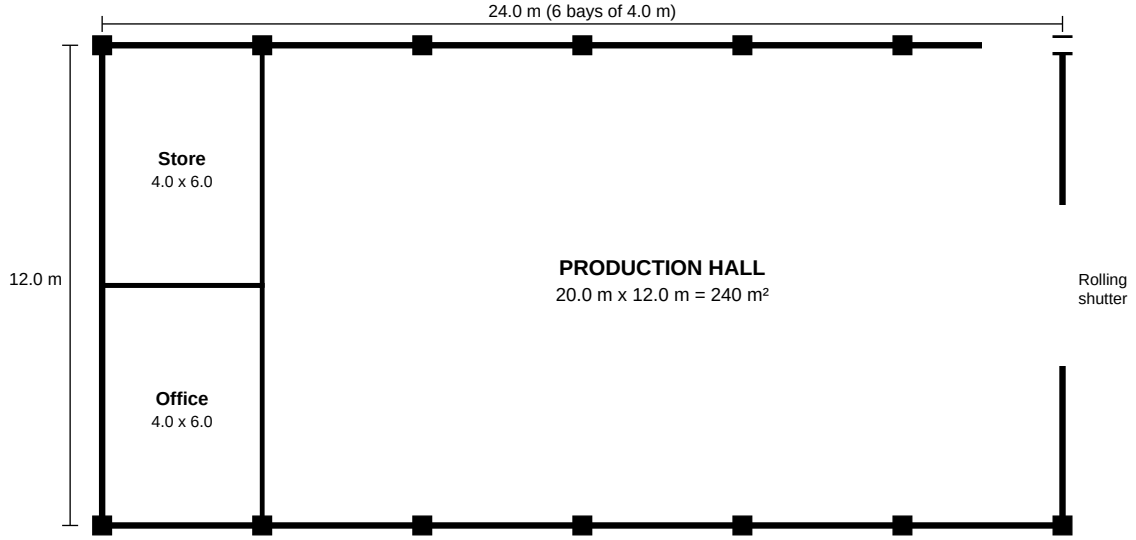


Fig: Plan of the industrial building (scale 1:100)

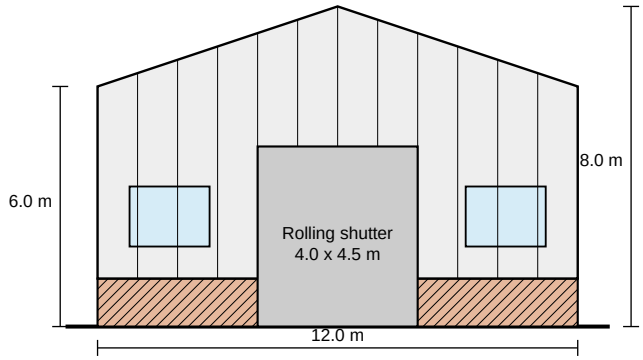


Fig: End elevation

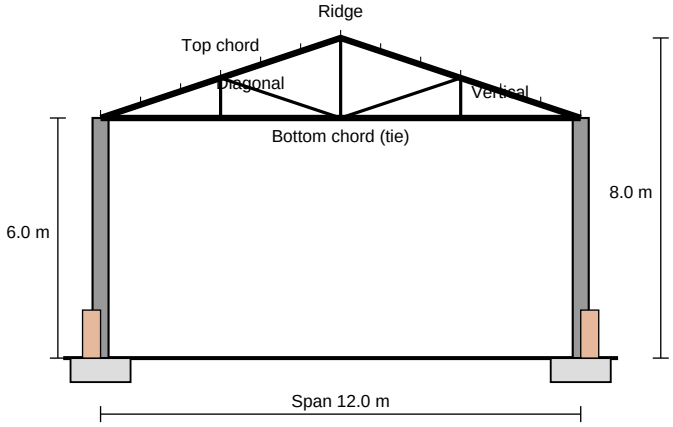


Fig: Section A-A (steel truss with columns)

Result:

Plan, elevation and section of the industrial shed of 24 m x 12 m were drawn.